Mo_2CT_x MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

Mo_2CT_x is a two-dimensional transition-metal carbide MXene prepared from a Mo_2Ga_2C precursor. This multilayer powder provides a layered, micron-scale research material for electrochemical, catalytic and interfacial studies. Surface terminations include –O, –OH, and –F, with distributions dependent on synthesis and lot.

KEY FEATURES

- Layered Mo_2CT_x morphology derived from Mo_2Ga_2C
- Micron-scale nominal flake / particle size
- Typical conductivity of 10–100 S/cm
- Mixed –O/–OH/–F surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Mo_2CT_x MXene
Precursor	Mo_2Ga_2C
Product form	Multilayer powder
Preparation	HF etching
Layer structure / count	Multilayer
Nominal flake / particle size	2–20 μm
Typical conductivity	10–100 S/cm
Surface terminations	–O, –OH, and –F
Appearance	Dark powder (descriptive only)
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

Multilayer Mo_2CT_x — SEM. Representative characterization; not lot-specific.

Product positioning: Supplied and technically supported by Winigen Materials for research and development use.